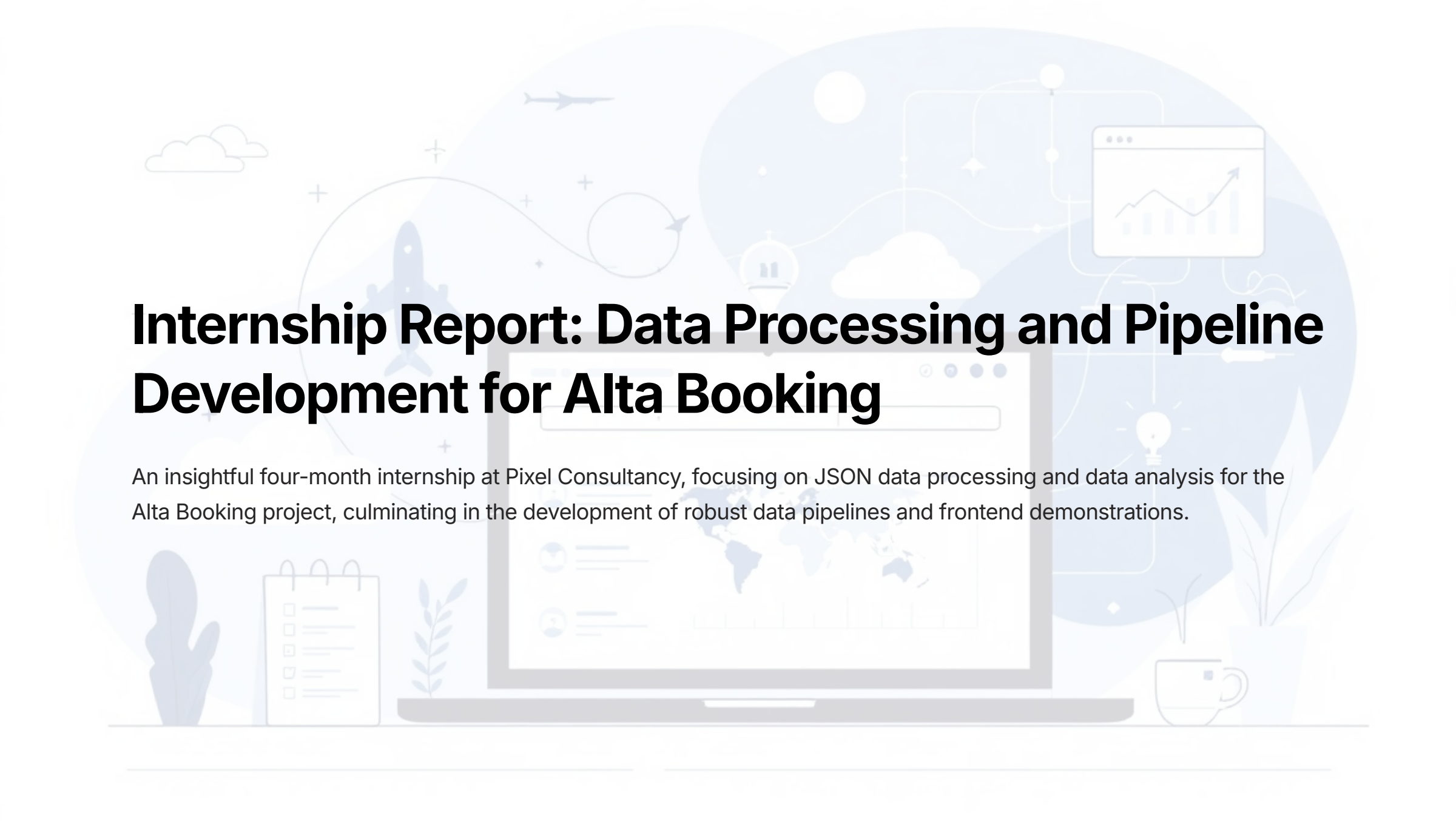




Internship Report: Data Processing and Pipeline Development for Alta Booking

An insightful four-month internship at Pixel Consultancy, focusing on JSON data processing and data analysis for the Alta Booking project, culminating in the development of robust data pipelines and frontend demonstrations.

Introduction to Pixel Consultancy and Alta Booking

Company Profile: Pixel Consultancy

Pixel Consultancy is an ISO-certified, 360° technology solutions provider based in Kolkata, India, offering web and mobile development, software products, digital marketing and AI/ML services. The company provides enterprise software solutions and consulting services across industries and has experience in building data-driven applications and analytics solutions.

Project/Company Summary: Alta Booking

Alta Booking is an online travel and lifestyle marketplace that aggregates travel services such as hotels, car rentals, trains, ferries and transfers, and offers AI-assisted trip planning tools. Alta Booking provides booking and itinerary services and acts as a marketplace connecting multiple travel service providers.

Background & Motivation

In the dynamic travel industry, Alta Booking must aggregate accurate, structured, and up-to-date data to remain competitive. The primary challenge is consolidating disparate travel data sources — often delivered in inconsistent JSON formats — into a unified, queryable dataset. Reliable, cleaned data is essential not only to display correct information to users (routes, pricing, availability) but also to train and power AI-assisted trip-planning tools and machine learning models that underpin personalized recommendations and pricing strategies. Without a robust pipeline for ingestion, validation, transformation, and storage, Alta Booking cannot guarantee data quality or fully leverage its analytics and AI capabilities, which would limit both user experience and business value.

Role and Responsibilities

- Ingestion and validation of diverse JSON data sources.
- Cleaning and structuring raw travel data for consistency.
- Database insertion and management within MongoDB.
- Preparation of clean datasets for machine learning model training.
- Implementation of front-end demonstration pages based on Figma designs.
- Coordination with clients to ensure data quality and schema adherence.

Objectives of the Internship

1

Develop a Scalable Data Ingestion Pipeline

To design and implement an efficient system capable of ingesting JSON data from various sources, including APIs and client files, ensuring data integrity and completeness.

2

Implement Robust Data Cleaning and Transformation

To create processes for validating, cleaning, and transforming raw JSON data into a standardised format suitable for downstream applications and analytical models.

3

Optimise Data Storage and Retrieval

To effectively utilise MongoDB for storing processed data, including schema design, indexing strategies, and ensuring efficient data retrieval for consumption.

4

Facilitate Machine Learning Data Preparation

To prepare high-quality, feature-engineered datasets tailored for training AI-assisted trip planning and recommendation models.

5

Contribute to Frontend Development

To translate Figma designs into functional, responsive web pages using HTML, CSS, and Bootstrap, providing visual demonstrations of processed data.

6

Enhance Client Data Workflow

To establish clear communication channels and documentation for managing client data submissions, schema changes, and reporting invalid records.

System Analysis and Design

High-Level Architecture

The data pipeline for Alta Booking was designed as a modular system, allowing for independent development and scalability of each component. Raw data from various sources, such as external APIs (e.g., Rome2Rio where permitted) and client-provided JSON files, enters the system through an ingestion layer. This raw data is then staged for initial processing, where schema validation and basic cleaning occur. A dedicated ETL process handles comprehensive data transformation, normalisation, and enrichment, preparing the data for storage. The cleaned and structured data is persistently stored in a NoSQL database (MongoDB). From MongoDB, data is made available for two primary consumers: the machine learning team for model training and the frontend application for display and user interaction. Logging and auditing mechanisms are integrated throughout the pipeline to monitor data quality and processing status.

Data Pipeline Flow

- **Sources:** Data originates from diverse sources including the Rome2Rio API (under explicit permission and licensing), other travel partner APIs, and direct JSON file uploads from clients.
- **Ingestion:** A Python-based script periodically fetches data from APIs or processes uploaded client files, storing the raw JSON in a staging area. This step includes initial error handling and logging.
- **Staging:** Raw JSON records are temporarily held in a designated area or collection within MongoDB (e.g., `raw_json` collection), awaiting detailed processing.
- **Cleaning/ETL:** A series of Python scripts, often leveraging libraries like pandas, performs extensive Extract, Transform, Load (ETL) operations. This includes flattening nested structures, normalising timestamps, unit conversions, deduplication based on unique identifiers, and data enrichment from auxiliary sources. Records failing validation are logged and flagged.
- **MongoDB Storage:** Cleaned and validated data is inserted into the primary MongoDB collection (e.g., `processed_records`), which is optimised with appropriate indexing for rapid queries. An `audits` collection tracks processing history.
- **Dataset Export for ML:** Specific scripts query the `processed_records` collection to generate feature-engineered datasets, which are then exported in formats like CSV or Parquet, ready for consumption by the machine learning team.
- **Frontend Consumption:** The frontend application directly queries the `processed_records` collection to retrieve and display travel information to users, enabling dynamic content presentation.

Implementation Details and Outcomes

1

JSON Ingestion & Schema Validation

JSON data for Alta Booking was sourced from multiple channels: licensed API endpoints like Rome2Rio, other travel service providers, and directly from clients as JSON files. The ingestion process involved reading these diverse inputs into a staging area. For schema validation, a conceptual approach was employed using predefined JSON schemas. This ensured that each incoming record adhered to the expected structure, mandating critical fields and data types. Records that failed validation were logged with detailed error messages, allowing for subsequent investigation and client feedback. The process aimed for early detection of malformed or incomplete data, preventing corruption of the main database.

2

ETL & Cleaning

The Extract, Transform, Load (ETL) phase was critical for standardising the heterogeneous data. Key steps included flattening nested JSON objects to create a more accessible structure, normalising timestamp formats to a consistent UTC standard, and performing unit conversions (e.g., currency, distance). Deduplication was implemented based on unique identifiers to prevent redundant entries. Data enrichment involved cross-referencing with reference data where available. Extensive quality checks were performed post-transformation to ensure accuracy and completeness. These operations were conceptually handled using data manipulation principles.

3

MongoDB Design & Insertion

MongoDB was chosen for its flexibility in handling semi-structured JSON data. Collections were structured into `raw_json` (for initial ingestion), `processed_records` (for clean, validated data), and `audits` (for tracking processing events and data quality metrics). To ensure data integrity, transactional-like patterns were followed, and idempotent inserts were implemented to prevent duplicate records upon reprocessing. Indexing strategies were designed to optimise common query patterns, such as searches by location, date ranges, or service type. Sample analytical queries were developed to demonstrate efficient data retrieval.

4

Preparing Data for Model Training

A significant outcome was the preparation of datasets for the machine learning team. This involved further feature engineering from the processed records, selecting relevant fields, and transforming them into formats optimal for model consumption, such as CSV or Parquet. Dataset versioning practices were adopted to manage changes over time, ensuring reproducibility of experiments. Datasets were packaged with metadata, describing their source, processing steps, and features, facilitating seamless handover to the ML teams. Approximately 20,000 JSON records were processed, cleaned, and organised, forming robust datasets for model training.